CS 598 Causal Methods for HCI — Long Homework 2

Due: 5:00 PM Central Time on May 4th. No late submissions are accepted.

Overview

This homework consists of five problems that span the core methods of the course: causal identification, Bayesian modeling, and principled estimation of treatment effects. The problems are grounded in plausible real-world contexts drawn from agriculture, education, anthropology, clinical trials, and regional policy—following the spirit of McElreath's *Statistical Rethinking* in using diverse empirical settings to teach general principles.

Problems 1 and 2 focus on causal identification and the challenges of estimating average treatment effects in realistic settings. Problems 3 and 4 address data quality issues: measurement error and missing data. Problem 5 is a challenge problem that combines spatial modeling with heterogeneous treatment effects.

Scoring note. Problems 1–3 will later be scaled to 25 points each, and Problems 4–5 will later be scaled to 15 points each.

General instructions.

- Your solutions must be typed and submitted as a PDF.
- Submit the homework on Gradescope, and make sure each page is properly assigned to the corresponding problem. Unassigned pages will not be graded.
- The questions are intentionally open-ended: you must exercise your own judgment about which modeling strategy is appropriate. There may be multiple valid approaches.
- For every model you fit, report MCMC diagnostics (R-hat, effective sample size, divergences) and visualize the posterior distributions with 89% HDI annotations.

Problem summary.

#	Title	Context	Key concepts
Q1	Confounded agricultural intervention	Fertilizer → crop yield	IV, colliders, mediation, counterfactuals
Q2	Teaching pedagogy and evaluations	Active learning → ratings	Ordered categorical, post-stratification
Q3	Wealth and educational attainment	Wealth → schooling	Measurement error, hierarchical shrinkage
Q4	Exercise and depression	Exercise → depression	M-DAG, missing data, imputation
Q5	Broadband and regional innovation	Broadband → patents	GP, Poisson, varying slopes (challenge)

1 The Confounded Agricultural Intervention (100 pts)

Context

An agricultural research institute is studying the effect of a new fertilizer program (X) on crop yield (Y) across 300 farms. They suspect that **farmer motivation** (U), which is **unobserved**, affects both fertilizer adoption and yield. They have data on:

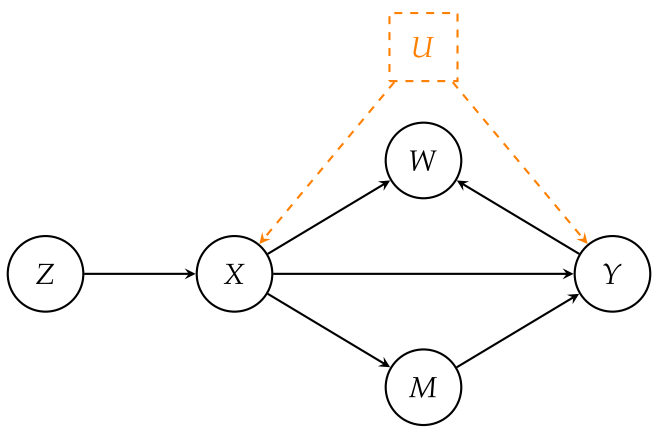
- **Soil quality** (Z): affects fertilizer adoption but not yield directly
- **Plant health** (M): lies on the causal path from fertilizer to yield
- **Farmer satisfaction** (W): caused by both fertilizer use and yield

Structural equations

$$\begin{aligned}
Z &\sim \mathcal{N}(0,1) \\
U &\sim \mathcal{N}(0,1) \quad \text{(unobserved)} \\
X &= \gamma\, Z + \alpha_X\, U + \epsilon_X \\
M &= \delta\, X + \epsilon_M \\
Y &= \beta_{\text{direct}}\, X + \lambda\, M + \alpha_Y\, U + \epsilon_Y \\
W &= \phi_X\, X + \phi_Y\, Y + \epsilon_W
\end{aligned} \tag{1}$$

where all error terms $\epsilon_\cdot \sim \mathcal{N}(0, 0.5^2)$ are mutually independent.

DAG



True parameter values (for verification)

Parameter	Symbol	True value
$Z \to X$	γ	0.7
$U \to X$	α_X	0.5
$X \to Y$ (direct)	β_{direct}	0.8
$X \to M$	δ	0.6
$M \to Y$	λ	0.5
$U \to Y$	α_Y	0.6
$X \to W$	ϕ_X	0.4
$Y \to W$	ϕ_Y	0.5
True total effect	$\beta_{\text{direct}} + \delta\lambda$	**1.1**
True direct effect	β_{direct}	0.8
True indirect effect	$\delta\lambda$	0.3

Questions

(a) DAG analysis (25 pts). Identify all paths from X to Y in the DAG above. Classify each path as causal or biasing. Determine whether there exists a valid adjustment set among the observed variables for estimating the *total* causal effect of X on Y. For each of the variables M and W, explain what would happen if you conditioned on it.

(b) Total causal effect estimation (25 pts). Estimate the total causal effect of X on Y. Verify that your estimate recovers a value close to the true total effect of 1.1. Explain why a naive regression of Y on X does not yield a valid causal estimate, and show this empirically.

(c) Effect decomposition (25 pts). Decompose the total causal effect into its direct and indirect components. Verify that your estimates are consistent with the true direct effect (0.8) and indirect effect (0.3). Discuss any challenges posed by the unobserved confounder for this decomposition.

(d) Individual counterfactual (25 pts). Select a specific farmer i who used fertilizer ($X_i > 0$) and observed yield Y_i. Compute the counterfactual: "What would this farmer's yield have been if they had not used fertilizer ($X_{\text{cf}} = 0$)?" Compare your counterfactual estimate with a posterior predictive estimate for a new farmer with $X = 0$, and explain the conceptual difference between the two.

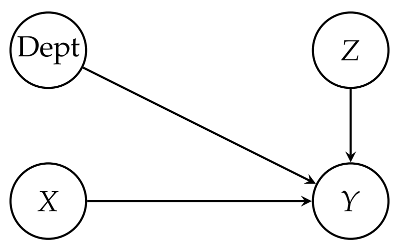
2 Teaching Pedagogy and Student Evaluations (100 pts)

Context

A large university is evaluating the effect of **active learning pedagogy** (X, binary: 0 = traditional lecture, 1 = active learning) on **end-of-semester course evaluation ratings** (Y, ordinal 1–5). Students are nested within **12 departments**, and we expect department-level variation in baseline ratings. **Student academic year** (Z, ordinal: 1 = freshman, 2 = sophomore, 3 = junior, 4 = senior) also affects ratings. The university wants to estimate the **overall Average Treatment Effect** of active learning, as well as **year-specific effects** for post-stratification.

There is **no confounding**—X is randomized within departments. Z is a pre-treatment covariate that improves precision.

DAG



Structural equations

$$
\begin{aligned}
\text{Rating}_i &\sim \text{OrderedLogistic}(\phi_i,\ \mathbf{c}) \\
\phi_i &= \beta_X X_i + \beta_Z \tilde{\delta}_{Z_i} + \alpha_{\text{dept}[i]} \\
\alpha_{\text{dept}} &= \sigma_{\text{dept}} \cdot z_{\text{dept}}, \quad z_{\text{dept}} \sim \mathcal{N}(0,1) \\
\tilde{\delta} &= \text{cumsum}(\boldsymbol{\delta}), \quad \boldsymbol{\delta} \sim \text{Dirichlet}(\mathbf{2})
\end{aligned} \tag{2}
$$

The **Dirichlet prior** on $\boldsymbol{\delta}$ enforces that year effects are monotonically non-decreasing: each successive year adds a non-negative increment. The total year effect is scaled by β_Z, so the sign and magnitude of the overall year trend is learned from the data. This is the standard approach for ordered categorical predictors (McElreath Ch. 12).

True parameter values (for verification)

Parameter	**Symbol**	**True value**
Treatment effect (log-odds)	β_X	0.8
Year effect scale	β_Z	0.6
Dirichlet simplex	$\boldsymbol{\delta}$	[0.3, 0.35, 0.35]
Cumulative year effects	$\beta_Z \tilde{\delta}$	[0, 0.18, 0.39, 0.6]
Department SD	σ_{dept}	0.4
Cutpoints	$\mathbf{c}$	[−1.5, −0.5, 0.5, 1.5]
Sample size	N	400
Departments		12

Questions

(a) Ordered categorical model (25 pts). Build an appropriate Bayesian model for this ordinal outcome that accounts for the ordered nature of both the outcome variable and the year predictor, as well as department-level variation in baseline ratings. Estimate the treatment effect on the log-odds scale and verify parameter recovery against the true values. Report MCMC diagnostics.

(b) ATE on the probability scale (25 pts). The log-odds coefficient β_X from your model is *not* the average treatment effect on the probability scale. Compute the ATE on the probability scale by appropriately averaging over the covariate distribution. Show how the treatment effect varies across rating thresholds by reporting exceedance probabilities $\Pr(Y \geq k)$ under treatment vs. control for each rating level k.

(c) Post-stratification by year (25 pts). Compute year-specific ATEs to examine how the probability-scale treatment effect varies across academic years. Then compute a population-weighted ATE using the known population proportions $[0.15, 0.25, 0.35, 0.25]$ for freshmen through seniors. Compare the post-stratified ATE to the overall ATE from part (b) and explain any differences.

(d) Comparison with a naive approach (25 pts). Fit a Bayesian linear regression that treats the ordinal rating as a continuous outcome. Compare the treatment effect estimate to the one from your ordinal model. Discuss the assumptions this naive approach makes about ordinal data and when those assumptions might lead to misleading conclusions.

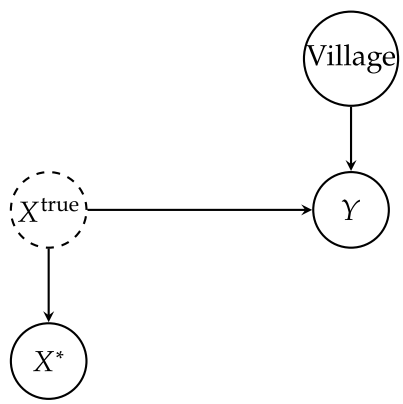
3 Household Wealth and Educational Attainment across Villages (100 pts)

Context

An anthropological study examines the relationship between household wealth and children's educational attainment across 15 villages in rural Tanzania. Household wealth is measured via self-report inventories (number of livestock, land area, assets), which are known to contain **substantial measurement error**. A separate validation study established that the **reliability** (correlation between test-retest) of the self-reported wealth measure is approximately **0.7**, corresponding to a measurement error standard deviation of $\sigma_{\text{me}} \approx 0.655$ (given that true wealth is standardized to unit variance).

True wealth affects educational attainment (years of schooling completed), and we expect **village-level variation** in baseline education levels. There is **no confounding** in this problem. The challenge is that we observe X^* (noisy reported wealth) rather than X^{true}.

DAG



Structural equations

$$
\begin{aligned}
X_i^{\text{true}} &\sim \text{Normal}(0, 1) \\
X_i^* &\sim \text{Normal}(X_i^{\text{true}},\ \sigma_{\text{me}}) \\
Y_i &= \alpha + \beta\, X_i^{\text{true}} + a_{\text{village}[i]} + \epsilon_i, \quad \epsilon_i \sim \text{Normal}(0, \sigma_Y)
\end{aligned}
\tag{3}
$$

where $a_{\text{village}[i]}$ are village-level varying intercepts drawn from a population distribution.

Attenuation bias

When we naively regress Y on X^* (ignoring measurement error), the coefficient is biased toward zero by a factor equal to the reliability:

$$\hat{\beta}_{\text{naive}} \approx \text{reliability} \times \beta_{\text{true}} = 0.7 \times 1.2 = 0.84$$

This is called **attenuation bias** (or regression dilution).

True parameter values (for verification)

Parameter	Symbol	True value
Wealth effect	β	1.2
Baseline education	α	6.0
Village-level SD	σ_{village}	1.5
Observation noise	σ_Y	1.0
Measurement error SD	σ_{me}	0.655
Reliability	ρ	≈ 0.7
Sample size	N	250
Villages		15

Questions

(a) Naive model (25 pts). Fit a hierarchical Bayesian regression of education (Y) on reported wealth (X^*), ignoring the measurement error. Include village-level varying intercepts. Report the posterior estimate of β and compare it to the true value of 1.2. Explain what you observe and why.

(b) Correcting for measurement error (25 pts). Build a model that accounts for the fact that reported wealth is a noisy measure of true wealth. Use the known measurement error information from the validation study. Show that your model recovers $\beta \approx 1.2$. Explain the key modeling idea that makes this correction possible.

(c) Model comparison (25 pts). Overlay the posterior distributions of β from both models. Quantify the magnitude of the attenuation bias and discuss the practical implications for the scientific conclusion about wealth and education.

(d) Hierarchical shrinkage (25 pts). Using your preferred model, visualize how the village-level intercept estimates are shrunk from their raw (unpooled) values toward the grand mean. Identify which villages are shrunk the most and explain why. Discuss the advantages of partial pooling over complete pooling or no pooling.

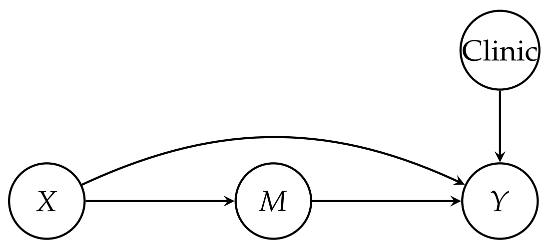
4 Exercise Intervention and Depression with Missing Engagement Data (100 pts)

Context

A clinical trial studies the effect of an exercise intervention (X, binary: 0 = control, 1 = exercise program) on depression scores (Y, continuous: Beck Depression Inventory, lower = less depressed) among 200 participants across 8 clinics. The study also collected weekly engagement scores (M, continuous), which mediates the effect of the intervention on depression—the exercise program works partly through increasing physical engagement. However, engagement scores are **missing for some participants**. Two scenarios are considered:

- **Scenario A (MAR):** Participants in the control group were less likely to report engagement scores (they found the tracking boring). Missingness depends on X (treatment assignment), which is observed.
- **Scenario B (MNAR):** Participants who had low engagement stopped reporting their scores (embarrassment/disengagement). Missingness depends on M itself (the missing variable).

DAG



The missingness indicator R_M ($R_M = 1$ if M is observed) has different parents depending on the scenario:

- Scenario A: $X \rightarrow R_M$
- Scenario B: $M \rightarrow R_M$

Structural equations

$$
\begin{aligned}
M_i &= \delta\, X_i + \epsilon_{M,i}, \quad \epsilon_{M,i} \sim \text{Normal}(0, \sigma_M) \\
Y_i &= \alpha + \beta_{\text{direct}}\, X_i + \lambda\, M_i + a_{\text{clinic}[i]} + \epsilon_{Y,i}, \quad \epsilon_{Y,i} \sim \text{Normal}(0, \sigma_Y) \\
\text{Total effect} &= \beta_{\text{direct}} + \delta \cdot \lambda
\end{aligned} \tag{4}
$$

True parameter values (for verification)

Parameter	Symbol	True value
Total effect	β_{total}	-3.0
Direct effect ($X \to Y$)	β_{direct}	-1.5
Exercise on engagement	δ	2.0
Engagement on depression	λ	-0.75
Indirect effect	$\delta \cdot \lambda$	-1.5
Baseline depression	α	20.0
Clinic intercept SD	σ_{clinic}	2.0
Engagement noise	σ_M	2.0
Depression noise	σ_Y	3.0
Sample size	N	200
Clinics		8

Questions

(a) M-DAG construction (25 pts). Draw the M-DAG (missingness DAG) for both Scenario A and Scenario B. Include the missingness indicator R_M and the edges that cause missingness. Classify the missingness mechanism in each scenario as MCAR, MAR, or MNAR. Explain what each mechanism means and why the distinction matters for downstream analysis.

(b) Total causal effect (25 pts). Estimate the total causal effect of the exercise intervention on depression. Think carefully about which variables should and should not appear in your model for the total effect, and explain your reasoning. Your estimate should be valid regardless of what is happening with the engagement data.

(c) Effect decomposition under Scenario A (25 pts). Now decompose the total effect into direct and indirect components. This requires using the engagement variable M, which has missing values. Under Scenario A, build a model that appropriately handles the missing engagement data and estimates the direct effect (β_{direct}) and indirect effect ($\delta \cdot \lambda$). Verify that the decomposition is consistent with the total effect from part (b). Include clinic-level varying intercepts.

(d) Effect decomposition under Scenario B (25 pts). Apply the same modeling approach from part (c) to the Scenario B data. Compare the resulting estimates to the true parameter values. Explain why the results differ from Scenario A. Discuss what additional assumptions, data, or analyses would be needed to obtain valid direct effect estimates under this missingness mechanism.

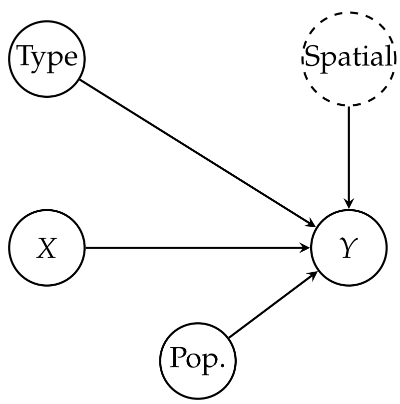
5 Broadband Access and Community Innovation across Regions — Challenge Problem (100 pts)

Context

A policy research institute studies the effect of broadband internet access on community innovation (measured by patent filings per year) across 15 regions. Nearby regions tend to have similar innovation rates due to shared economic conditions, labor markets, and knowledge spillovers—creating **spatial correlation** that must be modeled. The outcome is a **count variable** (patents per year), suggesting Poisson regression. Regions also vary by type (urban = 0, rural = 1), and the broadband effect may differ across region types.

A government broadband subsidy program was rolled out based on administrative criteria unrelated to geography or innovation—effectively a spatially randomized assignment. Region types (urban/rural) reflect population density classifications that are also assigned administratively, independent of regional coordinates.

DAG



X = broadband access (%), Y = patent count, Pop. = population (used as an exposure/offset), Type = urban/rural, Spatial = latent spatial effects modeled via a Gaussian process.

Model

$$\begin{aligned} Y_i &\sim \text{Poisson}(\mu_i) \\ \log(\mu_i) &= \alpha + \beta_{r[i]}\, X_i + k_i + \log(P_i) \\ \mathbf{k} &\sim \text{MVN}(\mathbf{0},\ \mathbf{K}) \\ K_{ij} &= \eta^2 \exp\!\left(-\rho^2\, d_{ij}^2\right) + \delta_{ij} \cdot 0.01 \end{aligned} \tag{5}$$

where $r[i]$ is the region type of region i, X_i is broadband access, P_i is population (used as an offset), and $\mathbf{k}$ is a vector of Gaussian process spatial effects with a squared-exponential kernel parameterized by amplitude η^2 and inverse length-scale ρ^2.

True parameter values (for verification)

Parameter	Symbol	True value
Baseline log-rate (per capita)	α	−7.0
Broadband effect, urban (log scale)	β_{urban}	2.0
Broadband effect, rural (log scale)	β_{rural}	1.5
GP max covariance	η^2	0.5
GP inverse length-scale	ρ^2	2.0
Number of regions	N	15

Interpretation of $\boldsymbol{\beta}$: $\beta_{\text{urban}} = 2.0$ means a 10 percentage-point increase in broadband access multiplies the per-capita patent rate by $\exp(0.10 \times 2.0) - 1 \approx 22\%$ for urban regions, and by $\exp(0.10 \times 1.5) - 1 \approx 16\%$ for rural regions.

Questions

(a) Naive Poisson regression (25 pts). Fit a Poisson regression of patent counts on broadband access with log(population) as an offset, ignoring spatial correlation. Examine the residuals for evidence of spatial structure. Discuss what consequences ignoring spatial correlation has for the credible intervals of the broadband effect.

(b) Spatial model (25 pts). Extend your model to account for spatial correlation between regions using an appropriate spatial prior over the distance matrix. Estimate the broadband effect and compare the posterior to the naive model from part (a). Explain why the credible intervals differ and which model you would trust more.

(c) Heterogeneous effects by region type (25 pts). Further extend the spatial model to allow the broadband effect to differ between urban and rural regions. Estimate the region-type-specific broadband effects and verify recovery against the true values. Quantify the evidence for heterogeneous treatment effects (i.e., is the urban effect credibly different from the rural effect?).

(d) ATE on the count scale (25 pts). The log-scale coefficient β is not directly interpretable as the average treatment effect on patent counts. Compute the ATE on the count scale for a 10 percentage-point increase in broadband access by appropriately marginalizing over the other model components. Explain why a naive back-transformation of the log-scale coefficient does not give the correct ATE, and demonstrate this empirically.

Homework 2 Rubric

Scoring note. The homework is written in a 100-points-per-problem format for clarity. As noted in the assignment overview, Problems 1–3 will later be scaled to 25 points each, and Problems 4–5 will later be scaled to 15 points each.

General expectations.

- Clearly identify the estimand before presenting a model or estimate.
- Report posterior summaries and the requested uncertainty intervals.
- Include the requested MCMC diagnostics where applicable.
- Justify modeling choices in causal terms, not only statistical terms.
- When comparing models, explain both the numerical difference and the substantive implication.

Problem 1: The Confounded Agricultural Intervention

Part	Raw Pts	A strong answer should include
(a) DAG analysis	25	All relevant paths from X to Y, with correct labels for causal, biasing, and collider-blocked paths. A correct statement about whether an observed adjustment set exists for the total effect. A clear explanation of why conditioning on M or W is inappropriate for the total effect.
(b) Total causal effect estimation	25	A valid causal strategy for the total effect under unobserved confounding, such as IV or an equivalent structural approach. A comparison showing why naive $Y \sim X$ regression is not causal here.
(c) Effect decomposition	25	A coherent decomposition of the total effect into direct and indirect components, with an explanation of why the additive decomposition works in this linear setting. Discussion of how the unobserved confounder complicates mediation.
(d) Individual counterfactual	25	An individual-level counterfactual for one observed treated farmer using abduction-action-prediction or an equivalent frozen-error construction. A clear distinction between this quantity and posterior prediction for a new untreated farmer.

Problem 2: Teaching Pedagogy and Evaluations

Part	Raw Pts	A strong answer should include
(a) Ordered categorical model	25	An ordered logistic model that respects the ordinal outcome, uses the monotonic year structure requested in the problem, and includes department-level variation. Clear interpretation of the treatment effect on the log-odds scale.
(b) ATE on the probability scale	25	A correct explanation that the log-odds coefficient is not itself the ATE on the probability scale. Marginalization over the covariate distribution when computing treatment effects on exceedance probabilities.
(c) Post-stratification by year	25	Year-specific treatment effects, correct use of the provided population weights, and an explanation of how the weighted result compares with the overall marginal ATE.
(d) Comparison with a naive approach	25	A Bayesian linear model treating the rating as continuous, compared directly to the ordinal model. A clear explanation of when this shortcut is misleading for ordinal outcomes.

Problem 3: Household Wealth and Educational Attainment across Villages

Part	Raw Pts	A strong answer should include
(a) Naive model	25	A hierarchical regression on reported wealth X^* with village-level variation, together with a clear explanation of why the slope is attenuated when measurement error is ignored.
(b) Correcting for measurement error	25	A latent-variable model that treats true wealth as unobserved and uses the validation-study information correctly. A clear explanation of how this model reduces attenuation bias.
(c) Model comparison	25	A direct comparison between the slope posteriors from the naive and measurement-error models, with a clear explanation of attenuation bias and why it matters substantively.
(d) Hierarchical shrinkage	25	A discussion of partial pooling at the village level, including which villages shrink the most and why. A comparison to complete pooling or no pooling strengthens the answer.

Problem 4: Exercise Intervention and Depression with Missing Engagement Data

Part	Raw Pts	A strong answer should include
(a) M-DAG construction	25	One M-DAG for each missingness scenario, correct classification of the mechanism in each case, and a clear explanation of why the distinction matters for identification and analysis.
(b) Total causal effect	25	A model for the total effect that does not condition on the mediator, with a correct explanation of why missing engagement does not prevent total-effect estimation here.
(c) Effect decomposition under Scenario A	25	A joint model for M and Y under MAR, principled handling of missing mediator values, and a decomposition into direct and indirect effects that is checked against the total effect.
(d) Effect decomposition under Scenario B	25	A clear explanation of why MNAR changes what is identifiable, why a naive MAR analysis becomes biased, and what additional assumptions, data, or sensitivity analysis would be needed.

Problem 5: Broadband Access and Community Innovation across Regions

Part	Raw Pts	A strong answer should include
(a) Naive Poisson regression	25	A Poisson model with the population offset, a check for spatial structure in the residual pattern, and a clear explanation of why ignoring spatial dependence affects uncertainty.
(b) Spatial model	25	A GP or equivalent spatial model over the distance structure, together with a comparison to the naive model and an explanation of which result is more credible and why.
(c) Heterogeneous effects by region type	25	A spatial model with distinct broadband effects for urban and rural regions, together with an assessment of whether the difference is credibly nonzero.
(d) ATE on the count scale	25	A count-scale ATE for a 10 percentage-point increase in broadband computed by marginalization rather than naive back-transformation, with a clear explanation of the difference between these quantities.